

HDOC - 33

Name	Gurveer Singh Cheema
SAF ID	590032925
Batch	15

Question

Write a C program to perform the following menu-driven operations using separate functions:

1. Read three variable-length 2D integer arrays A, B, and C, each of dimensions (m x n), and verify the associative property of matrix addition: $(A+B)+C = A+(B+C)$.
2. Read three variable-length 2D integer arrays with dimensions A (m x n), B (n x p), C (p x q). Verify the associative property of matrix multiplication: $(A \times B) \times C = A \times (B \times C)$.
3. Exit

Note: For each operation, display the matrices obtained on both sides of the equation and state whether the associative property is verified.

Program

```
#include <stdio.h>

void readM(int r,int c,int a[r][c]) {
    int i,j;
    for(i=0;i<r;i++)
        for(j=0;j<c;j++) scanf("%d",&a[i][j]);
}

void printM(int r,int c,int a[r][c]) {
    int i,j;
    for(i=0;i<r;i++) {
        for(j=0;j<c;j++) printf("%d ",a[i][j]);
        printf("\n");
    }
}

void add(int r,int c,int a[r][c],int b[r][c],int s[r][c]) {
    int i,j;
    for(i=0;i<r;i++)
        for(j=0;j<c;j++) s[i][j]=a[i][j]+b[i][j];
}

void multiply(int r1,int c1,int c2,int a[r1][c1],
              int b[c1][c2],int p[r1][c2]) {
    int i,j,k;
    for(i=0;i<r1;i++)
        for(j=0;j<c2;j++) {
            p[i][j]=0;
            for(k=0;k<c1;k++) p[i][j]+=a[i][k]*b[k][j];
        }
}

int same(int r,int c,int a[r][c],int b[r][c]) {
    int i,j;
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            if(a[i][j]!=b[i][j]) return 0;
    return 1;
}

int main() {
    int choice;
    do {
        printf("\n1. Associative property of addition\n");
        printf("2. Associative property of multiplication\n");
        printf("3. Exit\nEnter your choice: ");
        scanf("%d",&choice);

        if(choice==1) {
            int m,n,i;
            printf("Enter m and n: "); scanf("%d%d",&m,&n);
            int A[m][n],B[m][n],C[m][n];
            int AB[m][n],BC[m][n],L[m][n],R[m][n];

            printf("Enter A:\n"); readM(m,n,A);
            printf("Enter B:\n"); readM(m,n,B);
            printf("Enter C:\n"); readM(m,n,C);

            add(m,n,A,B,AB); add(m,n,AB,C,L);
            add(m,n,B,C,BC); add(m,n,A,BC,R);

            printf("(A+B)+C:\n"); printM(m,n,L);
```

```

        printf("A+(B+C):\n"); printM(m,n,R);

        if(same(m,n,L,R))
            printf("Associative property of addition verified.\n");
        else printf("Property not verified.\n");
    }
    else if(choice==2) {
        int m,n,p,q;
        printf("Enter m n p q: "); scanf("%d%d%d%d",&m,&n,&p,&q);
        int A[m][n],B[n][p],C[p][q];
        int AB[m][p],BC[n][q],L[m][q],R[m][q];

        printf("Enter A:\n"); readM(m,n,A);
        printf("Enter B:\n"); readM(n,p,B);
        printf("Enter C:\n"); readM(p,q,C);

        multiply(m,n,p,A,B,AB);
        multiply(m,p,q,AB,C,L);
        multiply(n,p,q,B,C,BC);
        multiply(m,n,q,A,BC,R);

        printf("(A*B)*C:\n"); printM(m,q,L);
        printf("A*(B*C):\n"); printM(m,q,R);

        if(same(m,q,L,R))
            printf("Associative property of multiplication verified.\n");
        else printf("Property not verified.\n");
    }
    else if(choice==3) printf("Exiting program...\n");
    else printf("Invalid choice\n");
} while(choice!=3);
return 0;
}

```

Sample Output

1. Associative property of addition
2. Associative property of multiplication
3. Exit

Enter your choice: 1

Enter m and n: 2 2

Enter A:

1 2

3 4

Enter B:

5 6

7 8

Enter C:

1 1

1 1

(A+B)+C:

7 9

11 13

A+(B+C):

7 9

11 13

Associative property of addition verified.

Enter your choice: 2

Enter m n p q: 2 2 2 2

Enter A:

1 2

3 4

Enter B:

5 6

7 8

Enter C:

1 0

0 1

(A*B)*C:

19 22

43 50

A*(B*C):

19 22

43 50

Associative property of multiplication verified.

Enter your choice: 3

Exiting program...

Result: The program successfully verifies the associative properties of matrix addition and matrix multiplication using separate functions and variable-length 2D arrays.